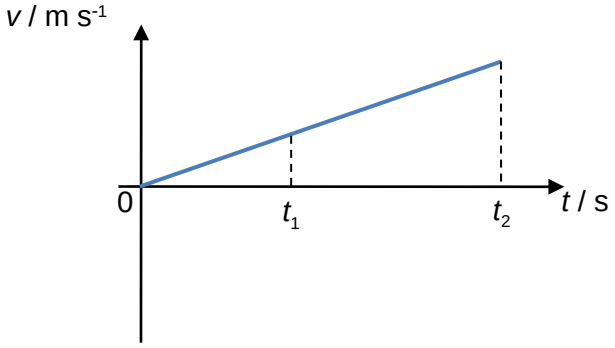
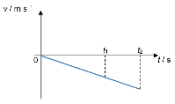
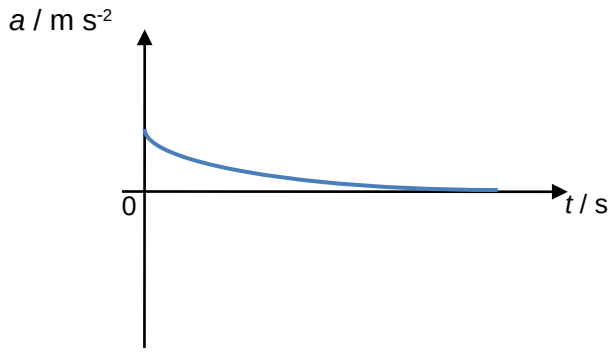
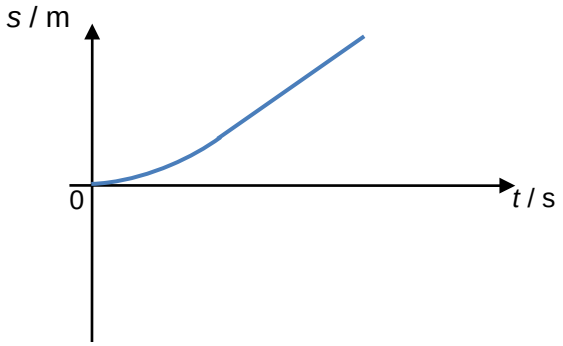
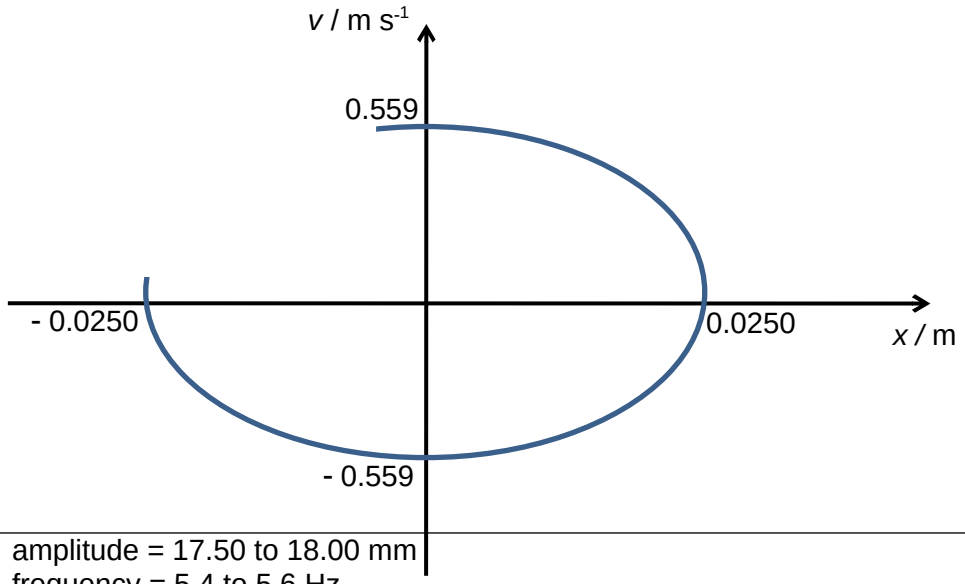
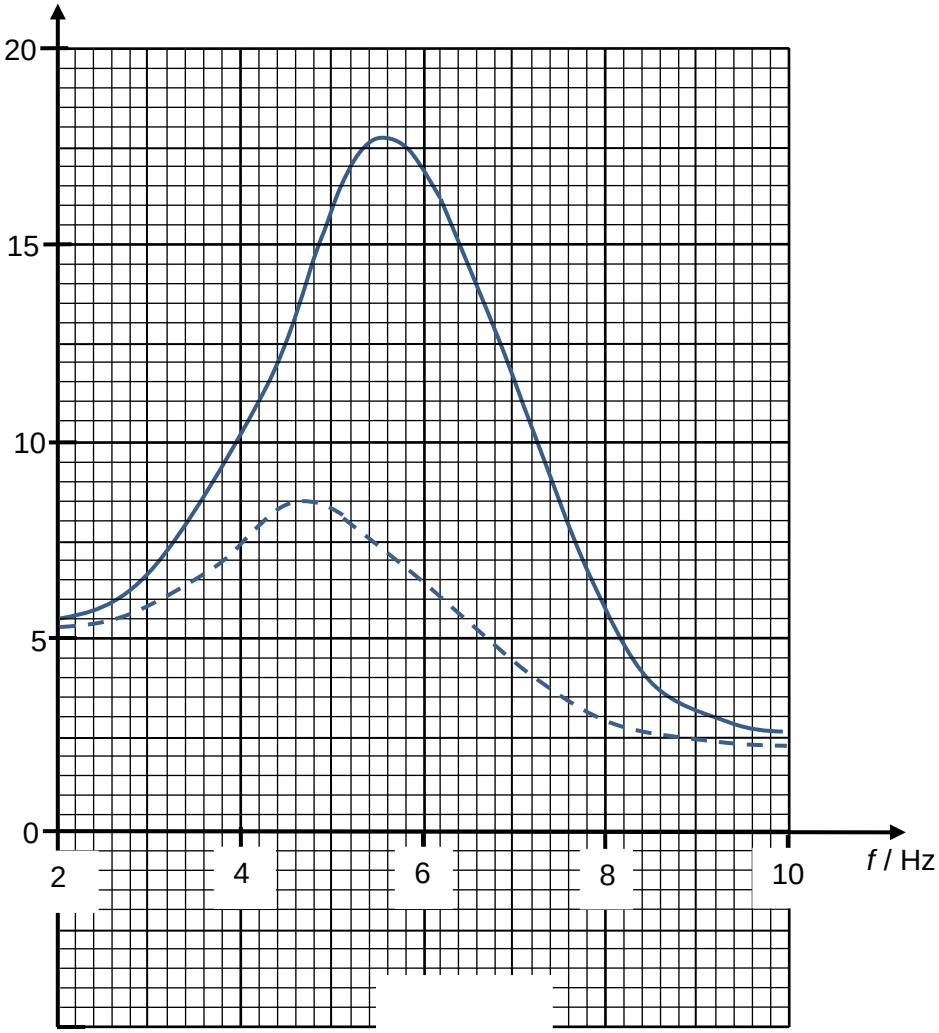
JC2 Preliminary Examinations Paper 2 Mark Scheme

Qn	Answer	Marks
1(a)(i)	component of weight parallel to surface $= mg\sin\theta$ $ma = mg\sin\theta$ $a = g\sin\theta$ $= 9.81\sin(35^\circ)$ $= 5.63\ \text{m s}^{-2}$	M1
1(a)(ii)	 Accept  too	A1
1(a)(iii)	area under v-t graph = distance travelled area under v-t graph $= \frac{1}{2}(u+v)t$ $= \frac{1}{2}((v - at)+v)t$ $= \frac{1}{2}(2.0 - (5.63)(0.30) + 2.0)(0.30)$ $= 0.3468 = 0.35\ \text{m (2sf)}$	M1 A1
1(a)(iv)	Since power is the rate of doing work OR $P=Fv$, Constant force + increasing distance covered per unit time (or increasing speed / velocity)	B1 B1
1(b)(i)	As speed increases, the <u>air resistance</u> opposing the block's motion <u>increases</u>. When magnitude of <u>air resistance</u> is <u>equal</u> to component of block's <u>weight</u> along the slope, there is <u>no net force</u> acting on the block. By Newton's first law of motion, this results in the block travelling at constant velocity.	B1 B1

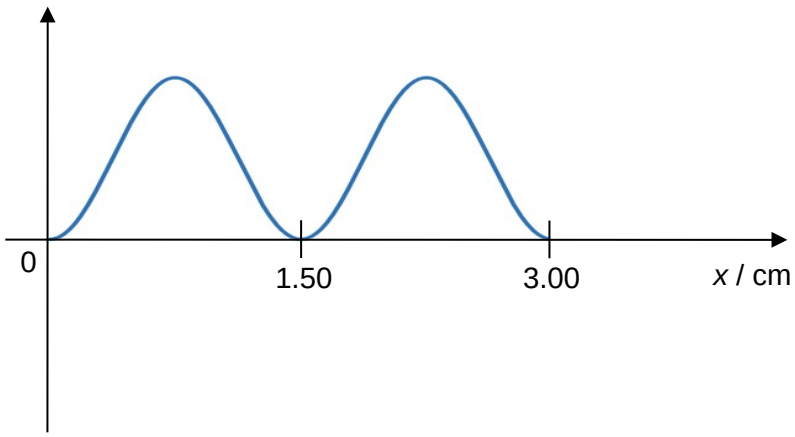
1(b)(ii)		A1
1(b)(iii)		A1
2(a)	$s_y = u_y t + \frac{1}{2} a t^2 \quad 0.018 = \frac{1}{2}(24) t^2$	M1
	$t = 3.87 \times 10^{-2}\ \text{s}$	A1
(b)	$p = v_x t$ $p = 2.9\ \text{cm}$	A1
(c)	$t = \frac{0.040}{0.75}$	M1
	$s = \frac{1}{2} a t^2 \quad s = \frac{1}{2}(24)\left(\frac{0.040}{0.75}\right)^2 = 0.0341$	M1
	minimum separation = 2s = 0.0682 m	A1
3(a)	Equal and opposite forces act on the two trolleys (Newton's Third Law) at every instant.	M1
	By Newton's 2nd law, rate of change of momentum will be the same for the two trolleys.	M1
	Since they act for the same time interval and have same force, the impulses (change in momentum) are equal in magnitude (and opposite in direction).	A1
(b)	No resultant external horizontal force acts on the system / Both trolleys experience the same change in momentum	M1
	Therefore the total momentum remains zero and hence the centre of gravity remains at rest.	A1

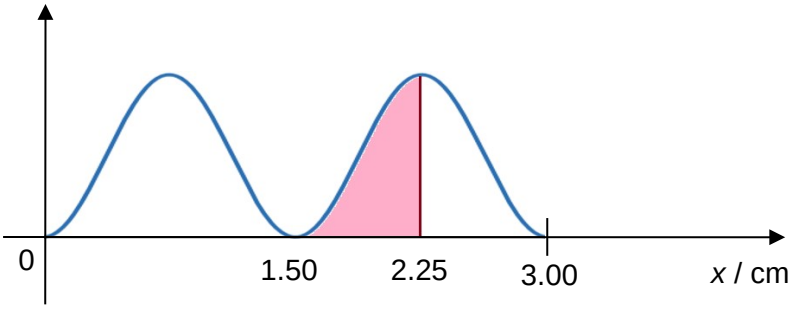
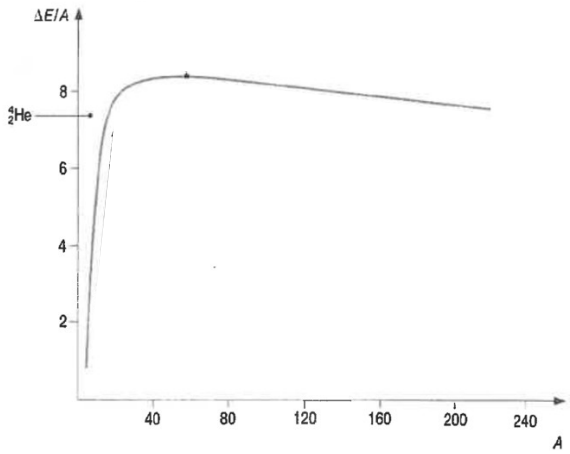
	By Newton's first law, the CG hence remains at rest.	
(c)	Applies conservation of momentum $M_1 v_1 = M_2 v_2$.	M1
	Uses $KE = \frac{1}{2} m v^2$ **and** substitutes/eliminates one velocity correctly. Alternatively, accept KE = p^2 / (2m) used in conjunction with the first equation.	M1
	Obtains $\frac{KE_A}{KE_B} = \frac{M_2}{M_1}$	A1
4(a)(i)	<u>The graph is a **straight** line that passes through the **origin**</u>. This indicates that the acceleration is **directly proportional** to the displacement, which is a characteristic of simple harmonic motion. <u>The gradient of the graph is **negative**</u>. This indicate that the direction of the acceleration and the displacement are in **opposing direction** **<u>or</u>** the acceleration is directed to a **fixed equilibrium position**.	B1 B1
(a)(ii)	From the graph, x_o = 2.50 cm = 0.025 m a_o = 12.50 m s^{-2} $a_o = \omega^2 x_o$	C1 (value of x_o)

	$\Rightarrow \omega^2 = 12.50 / 0.0250 = 500\ s^{-2}$ $v_o = x_o\omega = 0.559\ m\ s^{-1}$  **Fig. 4.3**	A1 (value of v_o) A1 (shape of graph)
(b)(i)	amplitude = 17.50 to 18.00 mm frequency = 5.4 to 5.6 Hz - both answers correct - both presented to correct d.p.	A1
(b)(ii)1.	W: Weight of ball T: Tension acting on the ball by the string  Note: T is upwards, W is downwards, and T must be larger than W	B1
(b)(ii)2.	At the maximum tension, the net force on the ball must be maximum. By Newton's second law of motion, $F_{net} = T_{max} - W$ $\therefore T_{max} = F_{net} + W$	C1

	$= ma_o + mg$ $= m(w^2x_o) + mg$ $= m(2\pi f)^2x_o + mg$ $= (0.150)[2\pi(5.6)]^2(0.0177) + (0.150)(9.81)$ $= 4.76$ N	M1 A1
(b)(iii)		B1 (similar shape: peak flatter) B1 (amp lower for all freq AND peak freq lower)
5(a)	A progressive wave is a wave in which <u>energy is carried from one point to another/ transfers energy</u> by means of <u>vibrations or oscillations</u> within the wave. <u>Particles</u> (or matter) within the wave are <u>not transported</u> along the wave.	B1
(b)	$\lambda = 4.0 \text{ cm} = 4.0 \times 10^{-2} \text{ m}$	A1
(c)	$v = 12.5 \times 4.0 \times 10^{-2} = 0.50 \text{ m s}^{-1}$ *Allow* ecf from (b)	A1
(d)	$\Delta x = \lambda \times \frac{\Delta \phi}{2\pi}$ $\Delta x = 4.0 \times \frac{5.03}{2\pi}$	

	$\Delta x = 3.2$ cm	B1
	P drawn correctly at 7.6 cm.	B1
(e)	Applies $I \propto A^2$ $\frac{I_2}{I_1} = \left(\frac{A_2}{A_1}\right)^2$.	M1
	¿ $\left(\frac{0.60}{0.90}\right)^2 = 0.44$	A1
6(a)	The e.m.f. induced in a conductor is directly proportional to the rate of change of magnetic flux linkage	B1
(b)(i)	Any two of t_1, t_3, t_5, t_7. Where gradient of the current-time graph is zero,	M1
	Therefore, rate of change of magnetic flux linkage is zero, so no e.m.f. is induced. $\frac{d(N\Phi)}{dt} = 0$,	A1
(b)(ii)	Between t_1 and t_2, the current in the solenoid is **decreasing.** Hence, the magnetic flux linkage through coil C is decreasing.	M1
	By **Lenz's law**, the induced magnetic field opposes this decrease by reinforcing the original magnetic field. Hence, it would be in the **same direction** as the original magnetic field.	A1
(c)	Area $A = \pi r^2 = \pi(1.8 \times 10^{-2})^2 = 1.018 \times 10^{-3}\,\text{m}^2$ Change in magnetic flux density $\Delta B = 9.4 \times 10^{-4} \times (5 - (-5)) = 9.4 \times 10^{-3}$ T	M1
	Change in flux linkage $\Delta(N\Phi) = NA\Delta B = 63(1.018 \times 10^{-3})(9.4 \times 10^{-3}) = 6.03 \times 10^{-4}$ Wb turns	M1
	Average induced e.m.f. $E_{\text{avg}} = \frac{\Delta(N\Phi)}{\Delta t} = \frac{6.03 \times 10^{-4}}{6.0 \times 10^{-3}} = 0.100$ V	A1
7(a)(i)	In the photoelectric experiment, electrons are observed to only be emitted when the surface is irradiated with <u>electromagnetic radiation</u> of frequency <u>above the threshold frequency</u>. This suggests that the <u>interaction</u> between an electron and a photon is <u>one-to-one</u>. The <u>absorption of **a** photon leads to a gain in energy for **an** electron</u> to be emitted.	B1 B1
7(a)(ii)	The formation of the interference pattern on the screen shows that electrons can <u>diffract through a slit</u>. Electrons can also undergo <u>constructive and destructive interference</u> to form regions of maxima and minima.	B1 B1

7(b)(i)	Allowed energy E for a particle in a box is given by $$E_n = \frac{h^2}{8mL^2}n^2$$ 'first excited state' $n = 2$ 'ground state' $n = 1$ Loss in energy of electron during transition from n=2 to n=1 $\Delta E_n = \frac{h^2}{8mL^2}(n_i^2 - n_f^2) = \frac{(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(0.0300)^2}(2^2 - 1^2)$ $= 2.01\times10^{-34}$ J Comments: A large number of students incorrectly stated that the energy of the electron at ground state is 0 J.	M1 A1
7(b)(ii)	$E = hf = \frac{hc}{\lambda}$ $2.01\times10^{-34} = \frac{(6.63\times10^{-34})(3.00\times10^{8})}{\lambda}$ $\therefore \lambda = 9.90\times10^{8}$ m	
7(b)(iii)1	/ cm^{-1}  	A1
7(b)(iii)2	area under graph = 1 $\left(\frac{A^2}{2}\right)(3.00) = 1$ $A = 0.816$ cm$^{-\frac{1}{2}}$	M1 A1

7(b)(iii)3	/ cm^{-1}  ∴ probability = 0.250	A1
8(a)(i)	mass defect $\Delta m = 2 \times 1.007825\,u + 2 \times 1.008665\,u - 4.002604\,u = 0.030376\,u$	M1
	binding energy per nucleon, $E = \Delta m c^2 = \dfrac{0.030376 \times 1.66 \times 10^{-27} \times \left(3.00 \times 10^8\right)^2}{1.60 \times 10^{-19} \times 10^6} \times \dfrac{1}{4} = 7.09$ MeV	A1
8(a)(ii)	 shape of graph. Upward sloping portion is about ¼ of graph with peak between *A* = 40 and *A* = 80. Downward sloping portion is about ¼ of upper portion of graph.	B1
8(a)(iii)	Product has higher binding energy per nucleon / more stable.	B1
8(a)(iv)	mass loss $\Delta m = 2.014102\,u + 1.007825\,u - 3.01603\,u = 0.005897\,u$ energy released $E = \Delta m c^2 = \dfrac{0.005897 \times 1.66 \times 10^{-27} \times \left(3.00 \times 10^8\right)^2}{1.60 \times 10^{-19} \times 10^6} = 5.51$ MeV	B1

8(b)	The decay of each particular parent nucleus releases the same amount of energy. and Some kinetic energy must have been carried away by another particle. Also accept: sum of the positrons' energy and neutrino's energy is constant for the decay of a particular nuclide	B1
	Positrons having varying kinetic energy implies have varying momentum (since $E = \frac{p^2}{2m}$) If there were only 2 particles, there would be a fixed momentum. (by principle of conservation of momentum)	B1
8(c)	Electric potential energy = $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r} = 8.99\times10^{9}\frac{\left(1.6\times10^{-19}\right)^2}{1.0\times10^{-14}} = 2.30144\times10^{-14}$ J	M1
	Using $E = \frac{3}{2}kT$, $2.30144\times10^{-14} = \frac{3}{2}\times1.38\times10^{-23}T$, giving $T = 1 \times 10^9$ K Also accept if students consider two proton approaching each other, hence uses $2\times\frac{3}{2}kT =$ electric potential energy, resulting in 0.6×10^9 K	A1
8(d)	6 – 2 = 4 protons	B1
8 (e)(i)	$\frac{0.1\times2.0\times10^{30}}{1.67\times10^{-27}}$ or $\frac{0.1\times2.0\times10^{30}}{1.007825\times1.66\times10^{-27}} = 1.1976\times10^{56}$	B1
(ii)	Energy = power × time $1.1976\times10^{56}\times6.68\times10^{6}\times1.6\times10^{-19} = 3.8\times10^{26}t$ giving $t = 1.07\times10^{10}$ years	B1